

Sri Lanka Institute of Information Technology



Lab Submission

IT25104066 – Dahanayaka D.K.S.G

Lab 01

Object Oriented Analysis and Design | SE2012

B.Sc. (Hons) Computer Science

Question 01: Running the First Java Program

```
import java.util.Scanner;

public class Welcome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter you first name: ");
        String firstName = sc.nextLine();
        System.out.print("Enter you last name: ");
        String lastName = sc.nextLine();
        System.out.printf("Welcome to the second year %s
%s\n",firstName, lastName);
    }
}
```

Output

```
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>javac Welcome.java
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>java Welcome
Enter you first name: Alex
Enter you last name: Jackson
Welcome to the second year Alex Jackson
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>|
```

Question 02

```
import java.util.Scanner;

public class Marks{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of students: ");
        int n = sc.nextInt();
        int marks[][] = new int[n][3];

        System.out.print("Enter:\n1:Add marks\n2:Get student
average\n3:Get subject average\n4:Get total mark of
student\n5:Exit\n");
        int choice = sc.nextInt();
        while (choice<1 || choice>5) {
            System.out.print("Invalid choice, enter again: ");
            choice = sc.nextInt();
        }

        while (true) {
            if (choice == 1) {
                System.out.println("Adding Mathematics marks:");
                for (int i=0; i<n; i++) {
                    System.out.printf("Student %d: ", i+1);
                    int inputMark = sc.nextInt();
                    marks[i][0] = inputMark;
                }
                System.out.println("Adding Chemistry marks:");
                for (int i=0; i<n; i++) {
                    System.out.printf("Student %d: ", i+1);
                    int inputMark = sc.nextInt();
                    marks[i][1] = inputMark;
                }
                System.out.println("Adding Physics marks:");
                for (int i=0; i<n; i++) {
                    System.out.printf("Student %d: ", i+1);
                    int inputMark = sc.nextInt();
                    marks[i][2] = inputMark;
                }
            } else if (choice == 2) {
                System.out.print("Enter student id to get the
student average: ");
                int stdId = sc.nextInt();
                while (stdId<1 || stdId>n) {
                    System.out.print("Invalid choice, enter again:
");
                    stdId = sc.nextInt();
                }
            }
        }
    }
}
```

```

        }
        float stdAvg = (marks[stdId-1][0] + marks[stdId-1][1] + marks[stdId-1][2]) / 3;
        System.out.printf("The average of student with student ID %d: %.1f\n", stdId, stdAvg);
    } else if (choice == 3) {
        System.out.println("Enter subject id to get the subject average:\n1:Mathematics\n2:Chemistry\n3:Physics");
        int subID = sc.nextInt();
        while (subID<1 || subID>3) {
            System.out.print("Invalid choice, enter again: ");
            subID = sc.nextInt();
        }
        float subAvg = 0;
        int subTotal = 0;
        for (int i=0; i<n; i++) {
            subTotal = subTotal + marks[i][subID-1];
        }
        subAvg = subTotal / n;
        System.out.printf("Average of this subject: %.1f\n", subAvg);

    } else if (choice == 4) {
        System.out.print("Enter student id to get the total mark of student: ");
        int stdId = sc.nextInt();
        while (stdId<1 || stdId>n) {
            System.out.print("Invalid choice, enter again: ");
            stdId = sc.nextInt();
        }
        int totalMark = marks[stdId-1][0] + marks[stdId-1][1] + marks[stdId-1][2];
        System.out.printf("The total mark of student with student ID %d: %d\n", stdId, totalMark);
    } else if (choice == 5) {
        System.out.println("Exiting...");
        break;
    }
    System.out.print("Enter:\n1:Add marks\n2:Get student average\n3:Get subject average\n4:Get total mark of student\n5:Exit\n");
    choice = sc.nextInt();
    while (choice<1 || choice>5) {
        System.out.print("Invalid choice, enter again: ");
        choice = sc.nextInt();
    }
}
}
}

```

Output

```
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>javac Marks.java
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>java Marks
Enter the number of students: 2
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Exit
1
Adding Mathematics marks:
Student 1: 67
Student 2: 80
Adding Chemistry marks:
Student 1: 78
Student 2: 90
Adding Physics marks:
Student 1: 45
Student 2: 66
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Exit
9
Invalid choice, enter again: 0
Invalid choice, enter again: 2
Enter student id to get the student average: 0
Invalid choice, enter again: 3
Invalid choice, enter again: 1
The average of student with student ID 1: 63.0
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Exit
3
```

```
Enter subject id to get the subject average:
1:Mathematics
2:Chemistry
3:Physics
6
Invalid choice, enter again: 2
Average of this subject: 84.0
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Exit
4
Enter student id to get the total mark of student: 0
Invalid choice, enter again: 2
The total mark of student with student ID 2: 236
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Exit
5
Exiting...

D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>|
```

Question 03

```
import java.util.Scanner;

public class MarksNew{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of students: ");
        int n = sc.nextInt();
        int marks[][] = new int[n][3];

        System.out.print("Enter:\n1:Add marks\n2:Get student
average\n3:Get subject average\n4:Get total mark of
student\n5:Display grade\n6:Exit\n");
        int choice = sc.nextInt();
        while (choice<1 || choice>6) {
            System.out.print("Invalid choice, enter again: ");
            choice = sc.nextInt();
        }

        while (true) {
            if (choice == 1) {
                System.out.println("Adding Mathematics marks:");
                for (int i=0; i<n; i++) {
                    System.out.printf("Student %d: ", i+1);
                    int inputMark = sc.nextInt();
                    marks[i][0] = inputMark;
                }
                System.out.println("Adding Chemistry marks:");
                for (int i=0; i<n; i++) {
                    System.out.printf("Student %d: ", i+1);
                    int inputMark = sc.nextInt();
                    marks[i][1] = inputMark;
                }
                System.out.println("Adding Physics marks:");
                for (int i=0; i<n; i++) {
                    System.out.printf("Student %d: ", i+1);
                    int inputMark = sc.nextInt();
                    marks[i][2] = inputMark;
                }
            } else if (choice == 2) {
                System.out.print("Enter student id to get the
student average: ");
                int stdId = sc.nextInt();
                while (stdId<1 || stdId>n) {
                    System.out.print("Invalid choice, enter again:
");
                    stdId = sc.nextInt();
                }
                float stdAvg = (marks[stdId-1][0] + marks[stdId-
1][1] + marks[stdId-1][2]) / 3;
            }
        }
    }
}
```

```

        System.out.printf("The average of student with
student ID %d: %.1f\n", stdId, stdAvg);
    } else if (choice == 3) {
        System.out.println("Enter subject id to get the
subject average:\n1:Mathematics\n2:Chemistry\n3:Physics");
        int subID = sc.nextInt();
        while (subID<1 || subID>3) {
            System.out.print("Invalid choice, enter again:
");

            subID = sc.nextInt();
        }
        float subAvg = 0;
        int subTotal = 0;
        for (int i=0; i<n; i++) {
            subTotal = subTotal + marks[i][subID-1];
        }
        subAvg = subTotal / n;
        System.out.printf("Average of this subject: %.1f\n",
subAvg);

    } else if (choice == 4) {
        System.out.print("Enter student id to get the total
mark of student: ");
        int stdId = sc.nextInt();
        while (stdId<1 || stdId>n) {
            System.out.print("Invalid choice, enter again:
");

            stdId = sc.nextInt();
        }
        int totalMark = marks[stdId-1][0] + marks[stdId-
1][1] + marks[stdId-1][2];
        System.out.printf("The total mark of student with
student ID %d: %d\n", stdId, totalMark);
    } else if (choice == 5) {
        System.out.print("Enter student id to display the
grade: ");
        int stdId = sc.nextInt();
        while (stdId<1 || stdId>n) {
            System.out.print("Invalid choice, enter again:
");

            stdId = sc.nextInt();
        }
        String grade = "";
        int stdMark = 0;
        for (int i=0; i<3; i++) {
            stdMark = marks[stdId-1][i];
            if (stdMark >= 90) {
                grade = "Grade A";
            } else if (stdMark >= 80) {
                grade = "Grade B";
            } else if (stdMark >= 70) {

```

```

        grade = "Grade C";
    } else if (stdMark >= 60) {
        grade = "Grade D";
    } else {
        grade = "Fail";
    }
    if (i==0) {
        System.out.printf("Mathematics: %s\n",
grade);
    } else if (i==1) {
        System.out.printf("Chemistry: %s\n", grade);
    } else {
        System.out.printf("Physics: %s\n", grade);
    }
}
} else if (choice == 6) {
    System.out.println("Exiting...");
    break;
}
System.out.print("Enter:\n1:Add marks\n2:Get student
average\n3:Get subject average\n4:Get total mark of
student\n5:Display grade\n6:Exit\n");
choice = sc.nextInt();
while (choice<1 || choice>6) {
    System.out.print("Invalid choice, enter again: ");
    choice = sc.nextInt();
}
}
}
}

```


Output

```
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>javac MarksNew.java
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>java MarksNew
Enter the number of students: 2
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Display grade
6:Exit
1
Adding Mathematics marks:
Student 1: 23
Student 2: 56
Adding Chemistry marks:
Student 1: 67
Student 2: 77
Adding Physics marks:
Student 1: 89
Student 2: 95
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Display grade
6:Exit
5
Enter student id to display the grade: 3
Invalid choice, enter again: 1
Mathematics: Fail
Chemistry: Grade D
Physics: Grade B
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Display grade
```

```
6:Exit
5
Enter student id to display the grade: 2
Mathematics: Fail
Chemistry: Grade C
Physics: Grade A
Enter:
1:Add marks
2:Get student average
3:Get subject average
4:Get total mark of student
5:Display grade
6:Exit
6
Exiting...
```

```
D:\MyFolder\SLIIT - Copy (2)\Y2S1\00AD\Lab\Lab01\SE2012-2025-S2-L01-IT25104066>|
```